

● MEDIUM

● ADVANCED MATH

● ANSWER KEY

Nonlinear equations in one variable and systems of equations in two variables

06

6 answers · Rationale-rich · Teach from doc

Q1**c****C. Two**

Choice C is correct. It is given that the constants a and c are both positive; therefore, the graph of the given quadratic equation is a parabola that opens up with a vertex on the y -axis at a point below the x -axis. The graph of the second equation provided is a horizontal line that lies above the x -axis. A horizontal line above the x -axis will intersect a parabola that opens up and has a vertex below the x -axis in exactly two points.

Choices A, B, and D are incorrect and are the result of not understanding the relationships of the graphs of the two equations given. Choice A is incorrect because the two graphs intersect. Choice B is incorrect because in order for there to be only one intersection point, the horizontal line would have to intersect the parabola at the vertex, but the vertex is below the x -axis and the line is above the x -axis. Choice D is incorrect because a line cannot intersect a parabola in more than two points.

Q2**4**

The correct answer is 4. The left-hand side of the given equation can be factored as $7x + 8x - 4$. Therefore, the given equation, $7x^2 - 20x - 32 = 0$, can be written as $7x + 8x - 4 = 0$. Applying the zero product property to this equation yields $7x + 8 = 0$ and $x - 4 = 0$. Subtracting 8 from both sides of the equation $7x + 8 = 0$ yields $7x = -8$. Dividing both sides of this equation by 7 yields $x = -\frac{8}{7}$. Adding 4 to both sides of the equation $x - 4 = 0$ yields $x = 4$. Therefore, the two solutions to the given equation, $7x^2 - 20x - 32 = 0$, are $-\frac{8}{7}$ and 4. It follows that 4 is the positive solution to the given equation.

Q3**B****B. 0**

Choice B is correct. Applying the zero product property to the given equation yields $3x = 0$, $x - 4 = 0$, and $x + 5 = 0$. Dividing each side of the equation $3x = 0$ by 3 yields $x = 0$. Adding 4 to each side of the equation $x - 4 = 0$ yields $x = 4$. Subtracting 5 from each side of the equation $x + 5 = 0$ yields $x = -5$. Therefore, the solutions to the given equation are 0, 4, and -5. Thus, one of the solutions to the given equation is 0.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q4

B

B. Exactly three

Choice B is correct. The given equation can be rewritten as

$52(x + 4)(x^2 - 4x + 16)(x^2 - 9)(x^2 + 9) = 0$, which is equivalent to $52(x + 4)(x^2 - 4x + 16)(x - 3)(x + 3)(x^2 + 9) = 0$. Applying the zero product property in this equation yields $52 = 0$, $x + 4 = 0$, $x^2 - 4x + 16 = 0$, $x - 3 = 0$, $x + 3 = 0$, and $x^2 + 9 = 0$. The equation $52 = 0$ has no solution. Subtracting 4 from both sides of the equation $x + 4 = 0$ yields $x = -4$. Adding 3 to both sides of the equation $x - 3 = 0$ yields $x = 3$. Subtracting 3 from both sides of the equation $x + 3 = 0$ yields $x = -3$. The discriminant of a quadratic equation of the form $ax^2 + bx + c = 0$ is $b^2 - 4ac$. If the discriminant is positive, the equation has two distinct real solutions. If the discriminant is zero, the equation has one distinct real solution. If the discriminant is negative, the equation has no real solutions. In the equation $x^2 - 4x + 16 = 0$, $a = 1$, $b = -4$, and $c = 16$. Substituting these values in the equation for the discriminant, $b^2 - 4ac$, yields $(-4)^2 - 4(1)(16)$, or -48 . Since the discriminant is negative, this equation has no real solutions. In the equation $x^2 + 9 = 0$, $a = 1$, $b = 0$, and $c = 9$. Substituting these values in the equation for the discriminant, $b^2 - 4ac$, yields $0^2 - 4(1)(9)$, or -36 . Since the discriminant is negative, this equation has no real solutions. Therefore, the distinct real solutions to the given equation are $x = -4$, $x = 3$, and $x = -3$. Thus, the given equation has exactly three distinct real solutions.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q5**either 0 or 3**

The correct answer is either 0 or 3. For an ordered pair to satisfy a system of equations, both the x - and y -values of the ordered pair must satisfy each equation in the system. Both expressions on the right-hand side of the given equations are equal to y , therefore it follows that both expressions on the right-hand side of the equations are equal to each other:

$x^2 - 4x + 4 = 4 - x$. This equation can be rewritten as $x^2 - 3x = 0$, and then through factoring, the equation becomes $x(x - 3) = 0$. Because the product of the two factors is equal to 0, it can be concluded that either $x = 0$ or $x - 3 = 0$, or rather, $x = 0$ or $x = 3$. Note that 0 and 3 are examples of ways to enter a correct answer.

Q6**B**

B. $n = \frac{7m}{5} - p$

Choice B is correct. It's given that the equation $7m = 5n + p$ relates the positive numbers m , n , and p . Dividing both sides of the given equation by 5 yields $\frac{7m}{5} = n + p$. Subtracting p from both sides of this equation yields $\frac{7m}{5} - p = n$, or $n = \frac{7m}{5} - p$. It follows that the equation $n = \frac{7m}{5} - p$ correctly gives n in terms of m and p .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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